

Alex Lee

Senior Data Scientist | Applied AI & Machine Learning
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PROFESSIONAL SUMMARY

Senior Data Scientist building and operating production ML systems in regulated, high-stakes environments. Experienced across end-to-end ML development, MLOps, causal inference, and applied deep learning, with a track record of delivering measurable revenue, operational, and clinical impact at scale.

SKILLS

Programming: Python (PyTorch, scikit-learn, LightGBM, Transformers, PySpark), R, SQL

Data Science: End-to-end ML, time series, NLP, deep learning, anomaly detection, causal inference, experiment design

MLOps & Data: MLflow, Databricks, Azure Synapse, Spark, Git, CI/CD

GenAI: RAG, vector databases, multi-agent systems, LLM memory & safety, LoRA

Communication: Tableau, Power BI; executive-level communication

EXPERIENCE

Humana

Senior Data Scientist

Sep 2024 – Present

- Architected and owned enterprise forecasting and **risk modeling** systems over **multimillion-record** medical and pharmacy claims, informing **CMS** healthcare quality-based bonus performance and materially improving program efficiency and operational decision making.
- Designed decision-aligned intervention models by optimizing ranking objectives under outcome **value**, intervention **cost**, and operational **capacity constraints**, materially improving targeting efficiency and downstream **ROI**.
- Led the **modernization** of legacy modeling pipelines, **scaling** small-sample models to enterprise-scale data through **transfer learning** and improved validation, materially improving generalization, stability, and time-to-production.
- Operationalized model **interpretability** as a system capability, using SHAP-based analysis to surface **actionable** drivers of risk and opportunity, enabling **defensible** interventions and increasing stakeholder **trust** and **adoption**.
- Embedded fairness, bias, and **auditability** into model design, evaluating subgroup performance and explanation consistency to meet **CMS regulatory scrutiny** and **governance** requirements.
- Owned production ML systems end-to-end, including **batch** pipelines, reusable **feature stores**, **CI/CD**, automated retraining, performance monitoring, and governance for **20+** interdependent models in **regulated** environments.
- Provided **technical leadership** and **mentorship** to junior data scientists, guiding modeling strategy, validation rigor, and production **tradeoffs** to ensure scalable, reliable, and compliant ML systems.

Senior Business Intelligence Engineer

Jun 2023 – Sep 2024

- Owned the design and deployment of a **population-scale** fall-risk prediction system serving 5M Medicare members, selecting operating thresholds under **clinical capacity constraints** to double high-risk **precision** and deliver ~\$2M in annual value.
- Applied **causal** and **quasi-experimental** methods (propensity score matching, difference-in-differences) to evaluate intervention effectiveness, informing strategy and generating \$100K+ in annual cost savings and \$5M+ revenue.
- Led **7+ cross-functional experimentation** efforts from problem framing through deployment, influencing product, clinical, and operational decision-making.

The Hershey Company

Lead Business Intelligence Analyst

Dec 2022 – Jun 2023

- Built predictive and survival models to forecast employee turnover and talent pipeline risk, enabling targeted retention strategies and near real-time KPI monitoring through production **APIs**.

Senior Data Analyst

Mar 2022 – Dec 2022

- Conducted adverse impact analyses using (**Chi-square, t-tests, ANOVA**) to inform fair hiring decisions.
- Developed ROI models for tuition reimbursement programs, supporting \$6M+ in annual funding.

Data Analyst

Nov 2018 – Mar 2022

- Automated People Analytics **dashboards**, replacing third-party tools and saving \$200K annually.
- Optimized **ETL pipelines** and **data models**, reducing data prep time by 40 hours per month through SQL, and Alteryx.

EDUCATION

Master of Applied Data Science - University of Michigan, Ann Arbor

Dec 2026

Bachelor of Business Administration - University of North Texas

Aug 2015

PROJECT HIGHLIGHTS

- **Multi-Agent RAG Education System (Personal)**: Designed and implemented a multi-agent LLM system with planner and tutor agents, retrieval-augmented generation, short- and long-term memory, and safety-aware orchestration.
- [Algorithmic Stock Trading \(PyTorch\)](#): Built a Stock Market Prediction Algorithm combining technical analysis and deep learning (LSTM, Temporal Fusion Transformer) for multi-task forecasting of price changes and direction. Developed a modular pipeline covering data ingestion, preprocessing, model training, evaluation, and backtesting, with interactive notebooks for experimentation and detailed performance analysis.